



Trigonometry · Mark Scheme

Chapter 5: Trigonometry. M = method mark A = accuracy mark (depends on M) B = independent mark ft = follow through

Section A – Sine rule, cosine rule, area

- 1 $\frac{b}{\sin 70^\circ} = \frac{12}{\sin 50^\circ}$ M1; $b = 14.7$ cm A1.
- 2 $QR^2 = 25 + 64 - 2(5)(8)\cos 60^\circ$ M1 = 49 A1; $QR = 7$ cm A1.
- 3 $\frac{1}{2}(6)(10)\sin 150^\circ$ M1 = 15 cm² A1.
- 4 (a) Largest angle opposite 13 B1; $\cos C = \frac{49 + 64 - 169}{112} = -\frac{1}{2} \Rightarrow C = 120^\circ$ M1 A1. (b) $\frac{1}{2}(7)(8)\sin 120^\circ = 14\sqrt{3}$ cm² B1.
- 5(a) $\frac{\sin C}{10} = \frac{\sin 35^\circ}{7}$ M1; $\sin C = 0.8194$ A1; $C = 55.0^\circ$ or $C = 180^\circ - 55.0^\circ = 125.0^\circ$ A1. (Both valid since $125.0^\circ + 35^\circ < 180^\circ$.)
- 5(b) $A = 180^\circ - 35^\circ - 125.0^\circ = 20.0^\circ$ M1; $BC = \frac{7 \sin 20.0^\circ}{\sin 35^\circ} = 4.18$ cm A1.
- 5(c) $\frac{1}{2}(10)(4.18)\sin 35^\circ$ M1 = 12.0 cm² A1.

Section B – Radians

- 6 (a) $\frac{3\pi}{4}$ B1; (b) 72° B1.
- 7 Arc = $5(0.8) = 4$ cm B1; area = $\frac{1}{2}(25)(0.8) = 10$ cm² B1.
- 8 $\frac{1}{2}(36)\theta = 24 \Rightarrow \theta = \frac{4}{3}$ rad M1 A1; perimeter = $12 + 6(\frac{4}{3}) = 20$ cm A1.
- 9 $\frac{1}{2}(64)(\frac{\pi}{3} - \sin \frac{\pi}{3})$ M1 A1; $= \frac{32\pi}{3} - 16\sqrt{3}$ cm² A1.
- 10(a) $2r + r\theta = 20 \Rightarrow \theta = \frac{20 - 2r}{r}$ M1 A1; $A = \frac{1}{2}r^2 \cdot \frac{20 - 2r}{r} = 10r - r^2$ A1.
- 10(b) $A = 25 - (r - 5)^2$ (or differentiate) M1; max $A = 25$ cm² when $r = 5$ A1; $\theta = 2$ rad A1.
- 10(c) $\theta > 0 \Rightarrow r < 10$; $\theta < 2\pi \Rightarrow 20 - 2r < 2\pi r \Rightarrow r > \frac{10}{\pi + 1}$ M1; $\frac{10}{\pi + 1} < r < 10$ (i.e. $2.41 < r < 10$) A1.

Section C – Equations

- 11 60° B1, 300° B1.
- 12 Principal value -60° M1; $x = 120^\circ$ A1, 300° A1.
- 13 $0^\circ \leq 2x \leq 720^\circ$ M1; $2x = 60^\circ, 120^\circ, 420^\circ, 480^\circ$ A1; $x = 30^\circ, 60^\circ, 210^\circ, 240^\circ$ A1 A1.
- 14 $\cos(x - 20^\circ) = \frac{2}{3}$ M1; $x - 20^\circ = 48.2^\circ, 311.8^\circ$ A1; $x = 68.2^\circ, 331.8^\circ$ A1.
- 15(a) Sine curve shifted 45° left B1; $(0, \frac{\sqrt{2}}{2}), (135^\circ, 0), (315^\circ, 0)$ B1.
- 15(b) $\sin(x + 45^\circ) = \frac{1}{2}$, with $45^\circ \leq x + 45^\circ \leq 405^\circ$ M1; $x + 45^\circ = 150^\circ, 390^\circ$ A1; $x = 105^\circ, 345^\circ$ A1. Common error: including $x + 45^\circ = 30^\circ$.
- 15(c) $x + 45^\circ = 210^\circ, 330^\circ \Rightarrow x = 165^\circ, 285^\circ$ B1 B1.

Section D – $\sin^2 \theta + \cos^2 \theta = 1$

- 16 $\sin \theta = \frac{12}{13}$ B1; $\tan \theta = \frac{12}{5}$ B1.



- 17** $\cos^2 \theta = \frac{5}{9}$ M1; third quadrant so $\cos \theta = -\frac{\sqrt{5}}{3}$ A1; $\tan \theta = \frac{2}{\sqrt{5}} = \frac{2\sqrt{5}}{5}$ A1.
- 18** Expand both: $\sin^2 \theta + 2 \sin \theta \cos \theta + \cos^2 \theta + \sin^2 \theta - 2 \sin \theta \cos \theta + \cos^2 \theta$ M1; $= 2(\sin^2 \theta + \cos^2 \theta)$ A1 $= 2$ A1.
- 19** $2(1 - \sin^2 x) + 3 \sin x = 3$ M1; $2 \sin^2 x - 3 \sin x + 1 = 0$ A1; $(2 \sin x - 1)(\sin x - 1) = 0$ M1; $\sin x = \frac{1}{2} \Rightarrow x = 30^\circ, 150^\circ$ A1; $\sin x = 1 \Rightarrow x = 90^\circ$ A1.
- 20(a)** $3 \sin^2 x + 7 \sin x = 1 - \sin^2 x - 4$ M1; $4 \sin^2 x + 7 \sin x + 3 = 0$ A1.
- 20(b)** $(4 \sin x + 3)(\sin x + 1) = 0$ M1; $\sin x = -1 \Rightarrow x = 270^\circ$ A1; $\sin x = -\frac{3}{4} \Rightarrow x = 228.6^\circ, 311.4^\circ$ M1 A1.
- 20(c)** $2\theta = 228.6^\circ$ (smallest positive) M1; $\theta = 114.3^\circ$ A1.

Section E – Reciprocal functions

- 21** (a) $\sqrt{2}$ B1; (b) $\sqrt{3}$ B1.
- 22** $\sin x = -\frac{1}{2}$ M1; $x = 210^\circ, 330^\circ$ A1.
- 23** $\text{LHS} = \frac{1}{\cos \theta} - \cos \theta = \frac{1 - \cos^2 \theta}{\cos \theta}$ M1 $= \frac{\sin^2 \theta}{\cos \theta}$ A1 $= \sin \theta \cdot \frac{\sin \theta}{\cos \theta} = \sin \theta \tan \theta$ A1.
- 24** $2(\sec^2 x - 1) + \sec x = 1$ M1; $2 \sec^2 x + \sec x - 3 = 0$ A1; $(2 \sec x + 3)(\sec x - 1) = 0$ M1; $\sec x = 1 \Rightarrow x = 0^\circ, 360^\circ$ A1; $\cos x = -\frac{2}{3} \Rightarrow x = 131.8^\circ, 228.2^\circ$ A1.
- 25(a)** $1 + \cot^2 x - \cot x = 3$ M1; $\cot^2 x - \cot x - 2 = 0$ A1.
- 25(b)** $(\cot x - 2)(\cot x + 1) = 0$ M1; $\tan x = \frac{1}{2} \Rightarrow x = 0.464$ M1 A1; $\tan x = -1 \Rightarrow x = \frac{3\pi}{4} = 2.36$ A1.
- 25(c)** U-shapes/ $\cap$ -shapes with asymptotes $x = \frac{\pi}{2}, x = \frac{3\pi}{2}$ B1; turning point $(\pi, -1)$, end points $(0, 1), (2\pi, 1)$ B1.

Section F – Inverse trig functions

- 26** (a) $\frac{\pi}{4}$ B1; (b) $-\frac{\pi}{3}$ B1.
- 27** (a) $-\frac{\pi}{4}$ B1; (b) $\frac{5\pi}{6}$ B1. *Common error: a negative answer for arccos.*
- 28** Let $\theta = \arcsin \frac{1}{3}$, so $\sin \theta = \frac{1}{3}$ M1; $\cos \theta = \frac{2\sqrt{2}}{3}$ A1; $\tan \theta = \frac{1}{2\sqrt{2}} = \frac{\sqrt{2}}{4}$ A1.
- 29** (a) $2x - 1 = \frac{1}{2}$ M1; $x = \frac{3}{4}$ A1. (b) $-1 \leq 2x - 1 \leq 1 \Rightarrow 0 \leq x \leq 1$ B1.
- 30(a)** Decreasing curve from $(-1, \pi)$ to $(1, 0)$ B1; passes through $(0, \frac{\pi}{2})$ B1.
- 30(b)** $y = \arcsin x \Rightarrow x = \sin y = \cos(\frac{\pi}{2} - y)$ M1; since $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$, we have $0 \leq \frac{\pi}{2} - y \leq \pi$, so $\arccos x = \frac{\pi}{2} - y$ A1; hence $\arcsin x + \arccos x = \frac{\pi}{2}$ A1.
- 30(c)** $\frac{\pi}{2} - \arccos x = 2 \arccos x \Rightarrow \arccos x = \frac{\pi}{6}$ M1; $x = \frac{\sqrt{3}}{2}$ A1.

Section G – Addition formulae

- 31** $\cos(45^\circ - 30^\circ) = \cos 45^\circ \cos 30^\circ + \sin 45^\circ \sin 30^\circ$ M1; $= \frac{\sqrt{6} + \sqrt{2}}{4}$ A1.
- 32** $\tan(45^\circ + 30^\circ) = \frac{1 + \frac{1}{\sqrt{3}}}{1 - \frac{1}{\sqrt{3}}}$ M1 $= \frac{\sqrt{3} + 1}{\sqrt{3} - 1}$ M1 (*rationalise*) $= \frac{4 + 2\sqrt{3}}{2} = 2 + \sqrt{3}$ A1.
- 33** $\cos A = \frac{3}{5}, \sin B = \frac{12}{13}$ B1; (a) $\frac{4}{5} \cdot \frac{5}{13} + \frac{3}{5} \cdot \frac{12}{13} = \frac{56}{65}$ M1 A1; (b) $\frac{3}{5} \cdot \frac{5}{13} + \frac{4}{5} \cdot \frac{12}{13} = \frac{63}{65}$ A1.
- 34** $\frac{\sqrt{3}}{2} \sin x + \frac{1}{2} \cos x = 2 \cos x$ M1; $\frac{\sqrt{3}}{2} \sin x = \frac{3}{2} \cos x \Rightarrow \tan x = \sqrt{3}$ A1; $x = 60^\circ, 240^\circ$ A1 A1.
- 35(a)** $\cos \theta \cos 60^\circ - \sin \theta \sin 60^\circ + \sqrt{3} \sin \theta$ M1; $= \frac{1}{2} \cos \theta + \frac{\sqrt{3}}{2} \sin \theta$ A1; $= \sin \theta \cos 30^\circ + \cos \theta \sin 30^\circ = \sin(\theta + 30^\circ)$ A1.
- 35(b)** $\sin(\theta + 30^\circ) = \frac{1}{2}, 30^\circ \leq \theta + 30^\circ < 390^\circ$ M1; $\theta + 30^\circ = 30^\circ, 150^\circ$ A1; $\theta = 0^\circ, 120^\circ$ A1.
- 35(c)** Maximum 1 B1, at $\theta = 60^\circ$ B1.



Section H – Double-angle formulae

36 $2\left(\frac{1}{9}\right) - 1 \quad M1 = -\frac{7}{9} \quad A1.$

37 $\tan 2A = \frac{4}{1-4} = -\frac{4}{3} \quad M1 \quad A1; \sin A = \frac{2}{\sqrt{5}}, \cos A = \frac{1}{\sqrt{5}}, \sin 2A = \frac{4}{5} \quad B1.$

38 $\frac{1 - (1 - 2\sin^2 x)}{2\sin x \cos x} \quad M1 = \frac{2\sin^2 x}{2\sin x \cos x} \quad A1 = \tan x \quad A1.$

39 $1 - 2\sin^2 x = \sin x \quad M1; (2\sin x - 1)(\sin x + 1) = 0 \quad M1; x = 30^\circ, 150^\circ \quad A1; x = 270^\circ \quad A1.$

40(a) $\sin 2x \cos x + \cos 2x \sin x = 2\sin x \cos^2 x + (1 - 2\sin^2 x)\sin x \quad M1 \quad M1; = 2\sin x(1 - \sin^2 x) + \sin x - 2\sin^3 x = 3\sin x - 4\sin^3 x \quad A1.$

40(b) $3\sin x - 4\sin^3 x = 2\sin x \Rightarrow \sin x(1 - 4\sin^2 x) = 0 \quad M1; \sin x = 0 \Rightarrow x = 180^\circ \quad A1; \sin x = \pm \frac{1}{2} \Rightarrow x = 30^\circ, 150^\circ, 210^\circ, 330^\circ \quad A1 \quad A1.$

Common error: dividing by $\sin x$ and losing $x = 180^\circ$.

Section I – R form

41 $R = 13 \quad B1; \tan \alpha = \frac{12}{5} \quad M1; 13\sin(x + 67.38^\circ) \quad A1.$

42 $R\cos \alpha = \sqrt{3}, R\sin \alpha = 1 \quad M1; R = 2 \quad A1; \alpha = 30^\circ: 2\sin(x - 30^\circ) \quad A1.$

43 $R\cos \alpha = 2, R\sin \alpha = 3 \quad M1; R = \sqrt{13} \quad A1; \alpha = 56.3^\circ \quad A1.$

44 $\sin(x - 30^\circ) = \frac{1}{2} \quad M1; x - 30^\circ = 30^\circ, 150^\circ \quad A1; x = 60^\circ, 180^\circ \quad A1.$

45(a) $R = 17 \quad B1; \tan \alpha = \frac{15}{8} \quad M1; 17\sin(\theta + 61.93^\circ) \quad A1.$

45(b) $H = 25 - 17\sin(20t + 61.93^\circ); \text{max when } \sin(\dots) = -1 \quad M1; H_{\max} = 42 \text{ m} \quad A1.$

45(c) $20t + 61.93^\circ = 270^\circ \quad M1; t = 10.40 \quad A1; 10 \text{ minutes } 24 \text{ seconds} \quad A1.$

Section J – Small-angle approximations

46 $0.05 + \left(1 - \frac{0.05^2}{2}\right) \quad M1 = 1.04875 \quad A1.$

47 $\cos 2\theta \approx 1 - \frac{(2\theta)^2}{2} \quad M1 = 1 - 2\theta^2 \quad A1.$

48 $1 - \cos 3\theta \approx \frac{9}{2}\theta^2 \quad M1; \theta \sin \theta \approx \theta^2 \quad M1; \text{limit} \approx \frac{9}{2} \quad A1.$

49 $5\theta + 2\theta - (1 - 2\theta^2) \quad M1 \quad M1; = 2\theta^2 + 7\theta - 1 \quad A1.$

50(a) $1 - \cos 4\theta \approx 8\theta^2 \quad M1; 2\theta \sin 3\theta \approx 6\theta^2 \quad M1; \text{ratio} \frac{4}{3} \quad A1.$

50(b) Angle must be in radians $B1; 5^\circ = \frac{\pi}{36} \quad M1; 1 - \frac{1}{2}\left(\frac{\pi}{36}\right)^2 = 0.996192 \quad A1.$

50(c) $\cos 5^\circ = 0.996195 \quad M1; \text{percentage error} \approx 0.0002\% \quad A1.$

Roar-level challenge

51 $\frac{1 - (1 - 2\sin^2 \theta)}{1 + (2\cos^2 \theta - 1)} \quad M1; = \frac{2\sin^2 \theta}{2\cos^2 \theta} = \tan^2 \theta \quad A1; \tan \theta = \pm\sqrt{3} \quad M1; \theta = 60^\circ \quad A1, 120^\circ \quad A1.$